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**Java SE 27 & JDK 27**

# Interface Function<T,R>

**Type Parameters:**

T - the type of the input to the function

R - the type of the result of the function

**All Known Subinterfaces:**

UnaryOperator<T>

**Functional Interface:**

This is a functional interface and can therefore be used as the assignment target for a lambda expression or method reference.

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```
@FunctionalInterface
public interface Function<T,R>
```

Represents a function that accepts one argument and produces a result.

This is a functional interface whose functional method is apply(Object).

**Since:**

1.8

## *Method Summary*

| All Methods | Static Methods | |
|---|---|---|
| **Modifier and Type** | **Method** | **Description** |
| default <V> Function<T,V> | **andThen**(Function<? super R, ? extends V> after) | Returns a composed function that first applies this function to its input, and then applies the after function to the result. |
| R | **apply**(T t) | Applies this function to the given argument. |
| default <V> Function<V,R> | **compose**(Function<? super V, ? extends T> before) | Returns a composed function that first applies the before function to its input, and then applies this function to the result. |
| | **identity**() | |

| | |
|---|---|
| `static <T> Function <T,T>` | Returns a function that always returns its input argument. |

## *Method Details*

### apply

`R apply(T t)`

Applies this function to the given argument.

**Parameters:**

`t` - the function argument

**Returns:**

the function result

### compose

```
default <V> Function<V,R> compose
(Function<? super V, ? extends T> before)
```

Returns a composed function that first applies the `before` function to its input, and then applies this function to the result. If evaluation of either function throws an exception, it is relayed to the caller of the composed function.

**Type Parameters:**

`V` - the type of input to the `before` function, and to the composed function

**Parameters:**

`before` - the function to apply before this function is applied

**Returns:**

a composed function that first applies the `before` function and then applies this function

**Throws:**

`NullPointerException` - if before is null

**See Also:**

`andThen(Function)`

### andThen

```
default <V> Function<T,V> andThen
(Function<? super R, ? extends V> after)
```

Returns a composed function that first applies this function to its input, and then applies the `after` function to the result. If evaluation of either function throws an exception, it is relayed to the caller of the composed function.

**Type Parameters:**

`V` - the type of output of the `after` function, and of the composed function

**Parameters:**

`after` - the function to apply after this function is applied

**Returns:**

a composed function that first applies this function and then applies the `after` function

**Throws:**

`NullPointerException` - if after is null

**See Also:**

`compose(Function)`

## identity

```
static <T> Function<T,T> identity()
```

Returns a function that always returns its input argument.

**Type Parameters:**

`T` - the type of the input and output objects to the function

**Returns:**

a function that always returns its input argument

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Report a bug or suggest an enhancement
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